

SAT Hard Question 3

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Statistical Sampling and Margin of Error

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325°F . The research manager recorded the amount of time, in minutes, each oven takes to preheat to 325°F .

Based on the first sample, the manager estimated an average of 14.2 minutes, with a margin of error of 1 minute. Based on the second sample, the estimate was 14.4 minutes, with a margin of error of 2.3 minutes.

Assuming the margins of error were calculated the same way, which of the following best explains why the first sample had a smaller margin of error?

- A. The first sample contained fewer ovens than the second sample.
- B. The first sample contained more ovens than the second sample.
- C. The first sample took less time on average to preheat.
- D. The first sample took more time on average to preheat.

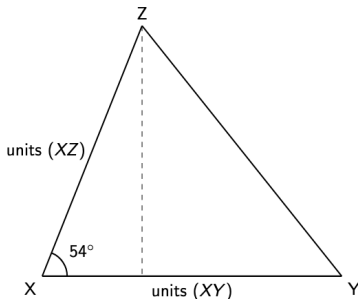
Solution: Margin of Error and Sample Size

Key Concept: The margin of error decreases with larger sample sizes (assuming similar variability).

- The first sample had a smaller margin of error (1 vs 2.3).
- This implies the first sample likely had a larger sample size.

Answer: B. The first sample contained more ovens than the second sample.

Area of Triangle Using Sine



In triangle XYZ , $\angle X = 54^\circ$, $XY = 26$ units, $XZ = 19$ units.
What is the area, in square units, of triangle XYZ ?

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

Solution: Triangle Area with Included Angle

Use the formula:

$$\text{Area} = \frac{1}{2}ab \sin C$$

Substitute:

$$\frac{1}{2} \cdot 26 \cdot 19 \cdot \sin 54^\circ = 247 \sin 54^\circ$$

Answer:

Question : Circle Geometry and Tangents

A circle with center O has radius 10. Point A lies on the circle, and chord AB subtends an angle of 120° at the center. A point P lies on the arc AB , and PA is extended to meet the tangent to the circle at B at point T . If $BT = x$, find the exact value of x .

(this question possesses some problems, so just skip that question)

- A. $10\sqrt{3}$
- B. 15
- C. 10
- D. 20

Answer to Question

We are given:

$$\angle AOB = 120^\circ, \quad r = 10.$$

Using the chord length formula:

$$AB = 2r \sin\left(\frac{\angle AOB}{2}\right) = 2 \cdot 10 \cdot \sin(60^\circ) = 20 \cdot \frac{\sqrt{3}}{2} = 10\sqrt{3}.$$

Now, applying geometry, since the triangle formed by the radius and the tangent is a right triangle:

$$OT = 20 \quad (\text{since } OT \text{ is the extension of radius } OP \text{ to the tangent})$$

$$BT^2 = OT^2 - OB^2 = 20^2 - 10^2 = 400 - 100 = 300.$$

$$BT = \sqrt{300} = 10\sqrt{3}.$$

Correct Answer: A. $10\sqrt{3}$

Question : Triangle in a Semicircle

In the figure, $\triangle ABC$ is inscribed in a semicircle with diameter $AC = 10$. Point B lies on the semicircle. If $AB = 6$ and $BC = x$, find x . Give your answer in simplest radical form.

- A. $\sqrt{36}$
- B. 8
- C. $\sqrt{44}$
- D. $\sqrt{64}$

Answer to Question

By Thales' theorem, any triangle inscribed in a semicircle is a right triangle with the right angle at B . Thus, apply the Pythagorean theorem:

$$AB^2 + BC^2 = AC^2.$$

$$6^2 + x^2 = 10^2 \quad \Rightarrow \quad 36 + x^2 = 100.$$

$$x^2 = 64 \quad \Rightarrow \quad x = 8.$$

Correct Answer: B. 8

Question : Intersection of a Parabola and a Circle

A parabola is given by

$$y = x^2 - 4x + 5,$$

and a circle by

$$x^2 + y^2 = 25.$$

The parabola and circle intersect at points P and Q . Find the distance between P and Q . (have P)

Answer Explanation

Solution Outline:

- Substitute the expression for y from the parabola into the circle's equation:

$$x^2 + (x^2 - 4x + 5)^2 = 25.$$

- Solve for x to find the x -coordinates of the intersection points P and Q .
- Substitute these x -values back into $y = x^2 - 4x + 5$ to obtain the corresponding y -coordinates.
- Use the distance formula between $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

After solving (details omitted for brevity), the distance is found to be:

3.88

Correct Answer: 3.88

Question (Be Careful!)

Solve the equation:

$$\sqrt{6r} = \sqrt{r^2 - 16}$$

What is the product of all solutions to the above equation?

Answer to Question

Step 1: Square both sides:

$$6r = r^2 - 16 \Rightarrow r^2 - 6r - 16 = 0$$

Step 2: Solve the quadratic:

$$r = \frac{6 \pm \sqrt{(-6)^2 + 4 \cdot 16}}{2} = \frac{6 \pm \sqrt{36 + 64}}{2} = \frac{6 \pm \sqrt{100}}{2} = \frac{6 \pm 10}{2}$$
$$\Rightarrow r = 8 \quad \text{or} \quad r = -2$$

Step 3: Check for extraneous solutions:

Plug into original:

$$\sqrt{6r} = \sqrt{r^2 - 16}$$

For $r = 8$:

$$\sqrt{48} = \sqrt{64 - 16} \Rightarrow \sqrt{48} = \sqrt{48} \quad \text{Valid}$$

For $r = -2$:

$$\sqrt{-12} = \sqrt{4 - 16} \Rightarrow \text{Not real} \quad (\text{Invalid})$$

Only valid solution: $r = 8$

Product of all valid solutions:

8